

LAC MEMPHREMAGOG, QC, Canada

WMO: 710359

Lat: 45.2677N

Lon: 72.1617W

Elev: 685

StdP: 14.34

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.6	-8.3	-21.2	1.8	-12.3	-16.2	2.3	-7.4	27.9	18.6	25.0	15.9	4.9	270	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	82.7	70.3	80.4	68.9	78.3	67.7	73.3	79.5	71.7	77.6	70.2	75.6	8.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.4	118.8	76.8	69.7	111.9	75.2	68.1	106.0	73.9	37.4	79.5	35.9	77.9	34.6	75.7	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.2	18.7	16.0		DB	-18.7	87.4	6.9	1.8	-23.6	88.7	-27.7	89.7	-31.5	90.7	-36.5	92.0
				WB	-20.2	75.8	7.5	2.2	-25.7	77.4	-30.1	78.6	-34.3	79.9	-39.8	81.4

Monthly Climatic Design Conditions

		Annual														
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	43.6	15.5	17.8	26.3	40.4	54.3	63.0	68.9	67.3	60.4	47.5	35.8	24.2	(6)
(7)		DBStd	20.73	13.43	11.81	11.30	8.37	7.45	6.24	4.96	4.98	7.13	8.31	9.04	11.03	(7)
(8)		HDD50	4439	1068	902	735	305	40	1	0	0	5	152	430	801	(8)
(9)		HDD65	8154	1533	1322	1199	737	340	111	18	34	174	545	875	1266	(9)
(10)		CDD50	2097	0	0	1	18	173	390	586	535	316	73	5	0	(10)
(11)		CDD65	337	0	0	0	0	7	50	139	105	35	1	0	0	(11)
(12)		CDH74	1815	0	0	0	4	62	274	781	521	171	2	0	0	(12)
(13)		CDH80	235	0	0	0	0	6	34	117	58	20	0	0	0	(13)
(14)	Wind	WSAvg	5.7	6.1	5.8	5.9	6.0	5.5	5.3	5.4	5.2	5.0	5.7	6.7	6.2	(14)
(15)	Precipitation	PrecAvg	46.5	3.5	2.6	3.1	3.4	3.8	4.5	5.2	4.6	4.0	4.3	3.7	3.9	(15)
(16)		PrecMax	55.7	6.1	3.7	4.5	7.4	6.2	6.7	8.9	7.4	8.5	9.2	5.6	8.0	(16)
(17)		PrecMin	35.4	1.7	1.4	1.5	0.4	1.9	2.8	2.2	1.3	1.9	1.7	1.8	2.0	(17)
(18)		PrecStd	5.1	1.1	0.7	0.7	1.7	1.1	1.2	1.6	1.5	1.7	2.0	1.0	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.1	47.9	58.6	71.0	80.1	83.9	86.2	84.6	82.9	72.9	63.3	50.4	(19)
(20)		MCWB	43.8	42.6	48.9	57.6	63.4	70.6	73.9	72.8	70.7	62.6	56.8	46.9	(20)	
(21)		2%	DB	41.2	41.5	50.0	64.7	75.0	80.2	83.2	81.4	78.4	68.2	57.1	45.8	(21)
(22)		MCWB	38.4	37.5	41.4	52.6	60.2	67.4	71.0	69.7	68.5	60.0	51.6	43.3	(22)	
(23)		5%	DB	37.9	38.1	45.1	59.0	71.2	77.2	80.7	79.2	75.4	64.2	52.8	42.0	(23)
(24)		MCWB	35.8	34.9	39.3	48.3	59.0	65.6	69.8	68.3	66.5	57.6	48.0	39.9	(24)	
(25)		10%	DB	34.5	34.9	41.2	53.8	67.3	74.4	78.3	76.9	72.4	60.5	48.9	38.2	(25)
(26)		MCWB	32.5	32.4	36.5	45.2	57.0	64.5	68.7	67.4	65.4	55.2	45.0	35.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.0	43.5	50.7	60.4	67.5	73.0	76.5	74.4	72.9	65.5	57.8	47.3	(27)
(28)		MCDWB	46.0	45.4	55.6	68.6	74.8	80.8	82.6	80.8	78.1	70.3	61.6	49.4	(28)	
(29)		2%	WB	39.2	38.4	43.3	54.8	64.4	70.8	74.2	72.3	70.8	61.9	52.8	43.6	(29)
(30)		MCDWB	40.5	40.8	47.7	62.9	71.0	76.9	79.8	78.5	76.3	65.7	56.9	45.4	(30)	
(31)		5%	WB	35.7	35.2	39.9	50.0	61.6	68.7	72.3	70.8	68.3	58.5	48.8	40.0	(31)
(32)		MCDWB	37.5	37.6	43.8	55.1	67.7	74.1	78.1	76.4	73.1	63.4	51.6	41.9	(32)	
(33)		10%	WB	32.6	32.7	37.1	46.5	58.9	66.5	70.6	69.3	66.1	55.6	45.2	36.2	(33)
(34)		MCDWB	34.6	34.7	40.9	52.2	64.6	71.7	76.2	74.5	71.3	59.8	48.6	37.8	(34)	
(35)	Mean Daily Temperature Range		MDBR	17.2	18.9	18.5	17.1	17.8	15.8	15.2	15.3	16.4	14.2	13.3	12.5	(35)
(36)		5% DB	MCDBR	17.3	18.8	22.9	26.9	23.7	19.9	17.6	18.4	19.3	19.9	19.6	13.2	(36)
(37)			MCWBR	16.3	15.4	15.9	15.3	11.8	9.9	8.8	9.1	10.7	12.5	15.4	12.8	(37)
(38)			5% WB	MCDWB	16.6	17.6	19.0	23.4	18.6	16.4	14.9	15.0	16.9	17.3	17.6	13.1
(39)		MCWBR		16.1	15.1	14.3	15.6	11.6	9.5	8.7	8.4	10.4	12.6	15.4	13.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.259	0.285	0.317	0.350	0.371	0.417	0.423	0.401	0.366	0.327	0.305	0.278	(40)	
(41)		taud	2.385	2.281	2.335	2.382	2.384	2.246	2.239	2.300	2.396	2.508	2.488	2.387	(41)	
(42)		Ebn at Noon	275	285	291	289	285	271	267	269	270	268	255	253	(42)	
(43)		Edh at Noon	23	31	34	36	37	43	43	38	32	25	21	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	450	738	1120	1372	1586	1748	1756	1563	1252	725	442	317	(44)	
(45)		RadStd	56	83	85	160	165	137	138	82	102	69	58	29	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)	
(47)	Regional (46 neighbors)	(m) N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	N/A	(47)

Nomenclature: See separate page